

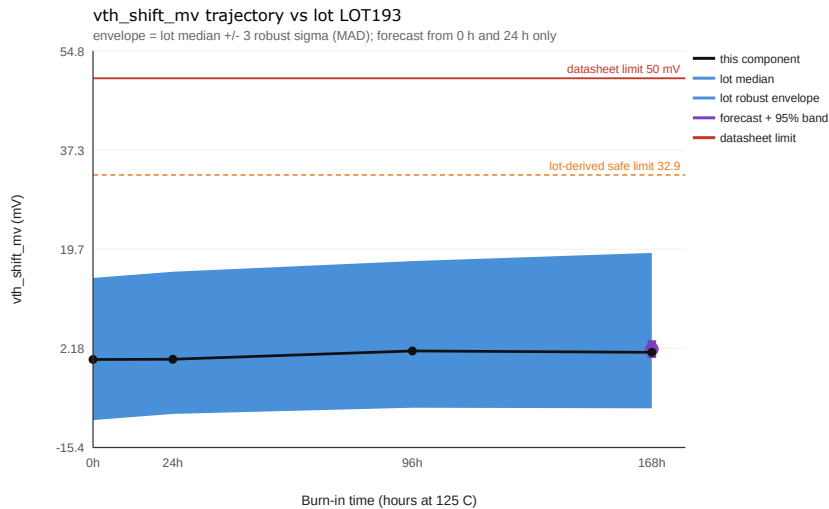
QA DISPOSITION REPORT burn-in parametric screening

Component	LOT193-C0488	Lot	LOT193	Board	LOT193-B07
Socket	r5 c0	Model version	dataset-v1.1 / ModuleA-d13e1ef1 / ModuleB-260260a2710:40		

DECISION: REVIEW

LAYER 1 — deterministic evidence (every value cites its comparison basis)

Absolute limit	PASS (97.00 nA vs 100.0 nA max, leakage_na)
Lot-relative level	+16.2 robust sigma vs lot median on leakage_na (median 22.64 nA, sigma 4.60 nA, n=500, MAD estimator; AEC-Q001 IQR/1.35 gives 4.80 nA)
Drift 0h to 24h	+0.043 mV on vth_shift_mv; lot median drift +1.014 mV, part is -1.2 robust sigma on the lot's drift distribution
Drift rate	0.0018 mV/h (lot median 0.0423 mV/h, 0.0x)
Predicted 168h	2.040 mV (vth_shift_mv, from 0h and 24h only)
95% upper bound	3.600 mV -> within lot-derived safe limit 32.879 mV
Joint anomaly	Mahalanobis D2 = 888.6 (good-part 99th pct = 16.8); largest contribution leakage_na at 52% of the total
Leakage/delay sign	leakage +16.2 sigma, delay -1.1 sigma -- consistent with the lot's process locus
Spatial check	no neighbour clustering on board LOT193-B07 (2/52 flagged), component-specific
Data quality	all checkpoints valid
Primary reason	elevated level relative to lot peers
Action	Hold. Repeat the measurement to rule out test error, then refer to a reliability engineer with this report.



LAYER 2 — model attribution

detector	normalised strength	fired	feature (Module B)	SHAP (mV)
L3a_MCD	67.70	YES	v24	-0.668
L3b_T2	28.03	YES	z:d1	+0.331
L4d_AutoEnc	24.08	YES	d1	-0.304
L4b_LOF	9.61	YES	lotmed_t0	-0.204
L1_dynamic_MAD	8.21	YES	lotmed_t24	+0.142

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

No single-parameter change clears the rules; the evidence is distributed across several measurements.

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

a defect present at incoming rather than one developing during burn-in [confidence: plausible]
Because: level is 16.2 robust sigma above the lot median at 0 h and the drift rate is unremarkable
Caveat: This is the problem statement's own worked example. Stable does not mean safe: the part is already an outlier at t=0, and burn-in will not reveal more about it. The right action is disposition on the level, not further stress.
Suggested next test: compare against incoming inspection data; FA if available

SIGN-OFF

Inspector		Date	
Disposition		Signature	
Notes			

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.